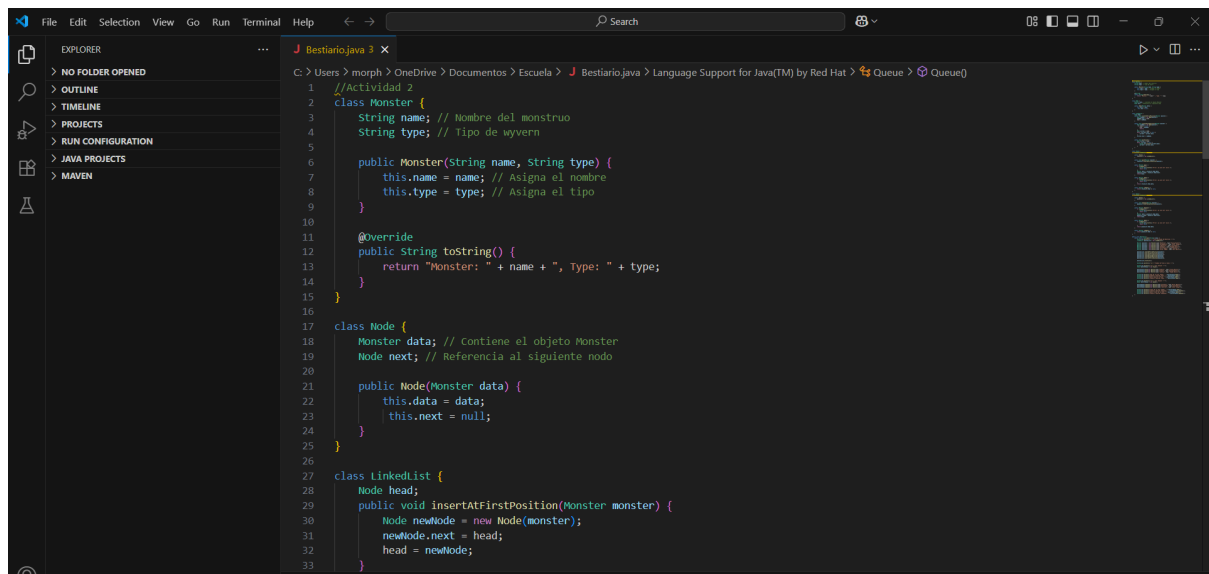


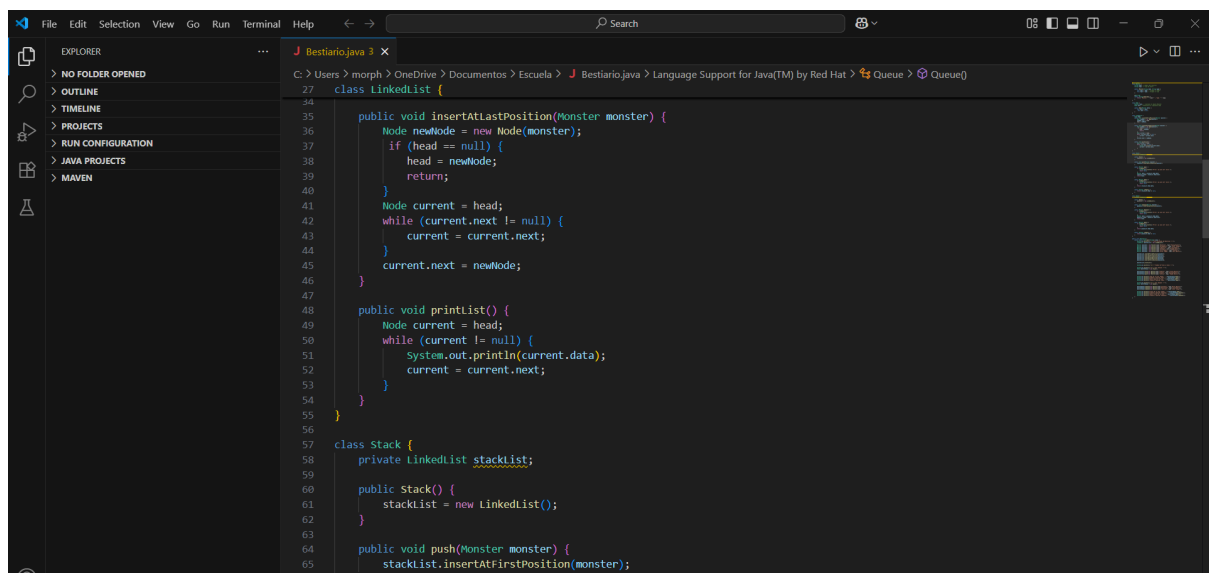
Nombre: Efren Aaron Gómez Casimiro	Matricula: AI07064860
Nombre del Curso: Estructura de Datos	Nombre del Profesor: Jesus Fernando Fabian Jocobi
Módulo: 3	Actividad:2
Fecha: 21/08/2025	



```

1 //Actividad 2
2 class Monster {
3     String name; // Nombre del monstruo
4     String type; // Tipo de wyvern
5
6     public Monster(String name, String type) {
7         this.name = name; // Asigna el nombre
8         this.type = type; // Asigna el tipo
9     }
10
11     @Override
12     public String toString() {
13         return "Monster: " + name + ", Type: " + type;
14     }
15 }
16
17 class Node {
18     Monster data; // Contiene el objeto Monster
19     Node next; // Referencia al siguiente nodo
20
21     public Node(Monster data) {
22         this.data = data;
23         this.next = null;
24     }
25 }
26
27 class LinkedList {
28     Node head;
29     public void insertAtFirstPosition(Monster monster) {
30         Node newNode = new Node(monster);
31         newNode.next = head;
32         head = newNode;
33     }

```



```

27 class LinkedList {
28
29     public void insertAtLastPosition(Monster monster) {
30         Node newNode = new Node(monster);
31         if (head == null) {
32             head = newNode;
33             return;
34         }
35         Node current = head;
36         while (current.next != null) {
37             current = current.next;
38         }
39         current.next = newNode;
40     }
41
42     public void printList() {
43         Node current = head;
44         while (current != null) {
45             System.out.println(current.data);
46             current = current.next;
47         }
48     }
49 }
50
51 class Stack {
52     private LinkedList stackList;
53
54     public Stack() {
55         stackList = new LinkedList();
56     }
57
58     public void push(Monster monster) {
59         stackList.insertAtFirstPosition(monster);
60     }

```

```
File Edit Selection View Go Run Terminal Help Search
J Bestiario.java X
C:\Users\morph>OneDrive>Documentos>Escuela>J Bestiario.java>Language Support for Java(TM) by Red Hat>Queue>Queue()

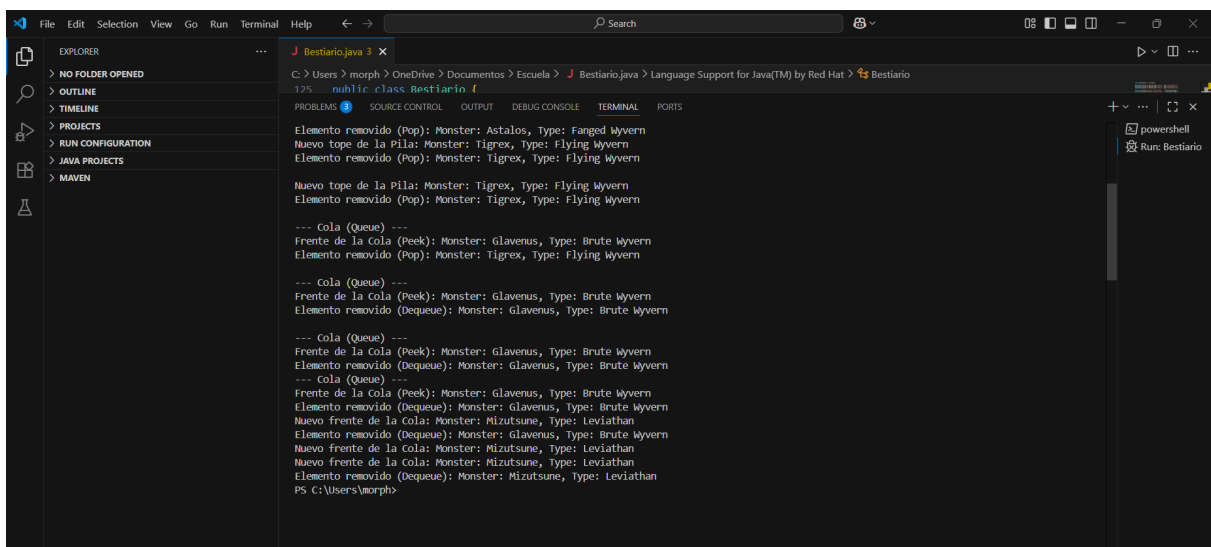
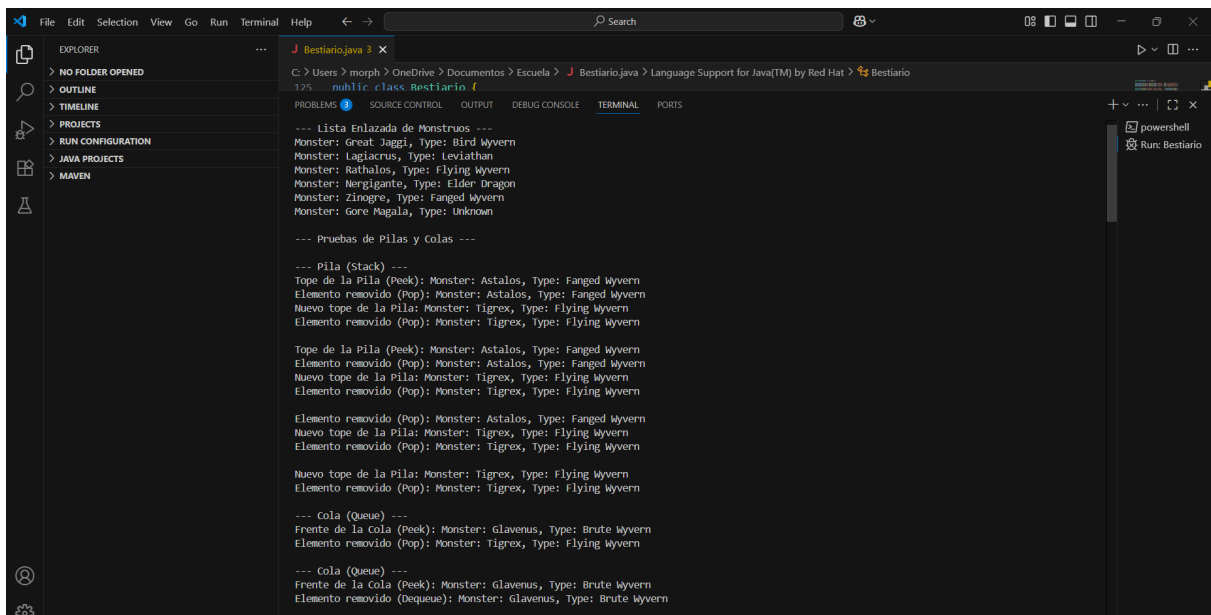
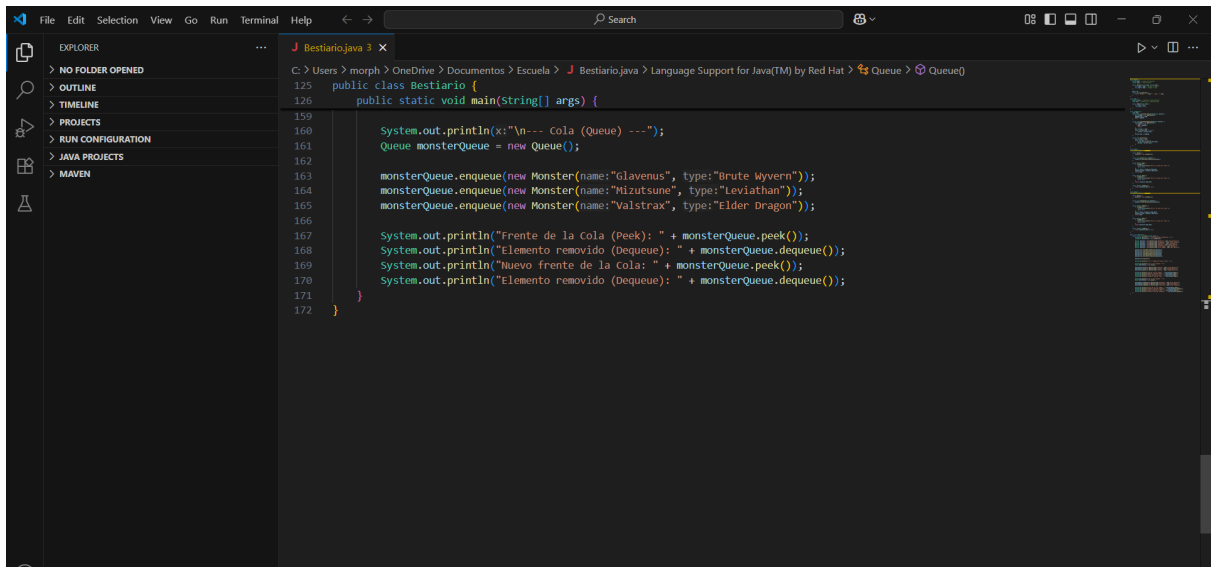
57 class Stack {
58     public Monster pop() {
59         if (isEmpty()) {
60             System.out.println(x:"Error: La pila está vacía.");
61             return null;
62         }
63         Monster data = stackList.head.data;
64         stackList.head = stackList.head.next;
65         return data;
66     }
67
68     public Monster peek() {
69         if (isEmpty()) {
70             System.out.println(x:"Error: La pila está vacía.");
71             return null;
72         }
73         return stackList.head.data;
74     }
75
76     public boolean isEmpty() {
77         return stackList.head == null;
78     }
79 }
80
81 class Queue {
82     private LinkedList queueList;
83
84     public Queue() {
85         queueList = new LinkedList();
86     }
87
88     public void enqueue(Monster monster) {
89         monster.setInsertAtAtPosition(monster);
90     }
91 }
```

```
File Edit Selection View Go Run Terminal Help Search
J Bestiario.java X
C:\Users\morph>OneDrive>Documentos>Escuela>J Bestiario.java>Language Support for Java(TM) by Red Hat>Queue>Queue()

91 class Queue {
92
93     public void enqueue(Monster monster) {
94         queueList.InsertAtAtPosition(monster);
95     }
96
97     public Monster dequeue() {
98         if (isEmpty()) {
99             System.out.println(x:"Error: La cola está vacía.");
100             return null;
101         }
102         Monster data = queueList.head.data;
103         queueList.head = queueList.head.next;
104         return data;
105     }
106
107     public Monster peek() {
108         if (isEmpty()) {
109             System.out.println(x:"Error: La cola está vacía.");
110             return null;
111         }
112         return queueList.head.data;
113     }
114
115     public boolean isEmpty() {
116         return queueList.head == null;
117     }
118 }
119
120 public class Bestiario {
121     Run [Debug] [Run main] [Debug main]
122     public static void main(String[] args) {
123         System.out.println(x:"--- Lista Enlazada de Monstruos ---");
124         LinkedList monsterList = new LinkedList();
125     }
126 }
```

```
File Edit Selection View Go Run Terminal Help Search
J Bestiario.java X
C:\Users\morph>OneDrive>Documentos>Escuela>J Bestiario.java>Language Support for Java(TM) by Red Hat>Queue>Queue()

125 public class Bestiario {
126     public static void main(String[] args) {
127         Monster monster1 = new Monster(name:"Rathalos", type:"Flying Wyvern");
128         Monster monster2 = new Monster(name:"Mergigante", type:"Elder Dragon");
129         Monster monster3 = new Monster(name:"Lagiacrus", type:"Leviathan");
130         Monster monster4 = new Monster(name:"Zinogre", type:"Fanged Wyvern");
131         Monster monster5 = new Monster(name:"Gore Magala", type:"Unknown");
132         Monster monster6 = new Monster(name:"Great Jaggi", type:"Bird Wyvern");
133
134         monsterList.InsertAtFirstPosition(monster1);
135         monsterList.InsertAtAtPosition(monster2);
136         monsterList.InsertAtFirstPosition(monster3);
137         monsterList.InsertAtAtPosition(monster4);
138         monsterList.InsertAtAtPosition(monster5);
139         monsterList.InsertAtFirstPosition(monster6);
140
141         monsterList.printtist();
142
143         System.out.println(x:"\n--- Pruebas de Pilas y Colas ---");
144
145         System.out.println(x:"\n--- Pila (Stack) ---");
146         Stack monsterStack = new Stack();
147
148         monsterStack.push(new Monster(name:"Diablos", type:"Flying Wyvern"));
149         monsterStack.push(new Monster(name:"Tigrex", type:"Flying Wyvern"));
150         monsterStack.push(new Monster(name:"Astalos", type:"Fanged Wyvern"));
151
152         System.out.println("Tope de la Pila (Peek): " + monsterStack.peek());
153         System.out.println("Elemento removido (Pop): " + monsterStack.pop());
154         System.out.println("Nuevo tope de la Pila: " + monsterStack.peek());
155         System.out.println("Elemento removido (Pop): " + monsterStack.pop());
156
157         System.out.println(x:"\n--- Cola (Queue) ---");
158     }
159 }
```



A continuación se presenta el código solicitado para la actividad con su respectivo funcionamiento con lo visto y practicado en clase.